INTERNSHIP PROJECT REPORT

On

C AND C++ DEVELOPMENT

Submitted by

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MALLA REDDY ENGINEERING COLLEGE

Submitted in partial fulfillment of the requirements

for the completion of internship

at

GROWFINIX

Under the Guidance of

Growfinix Team

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CERTIFICATE

This is to certify that Ms. V. MEENAKSHI has successfully completed the internship project report on **"C AND C++ DEVELOPMENT"** submitted to **Growfinix** during the internship period.

The work submitted by the student is genuine and completed successfully under the guidance and support of the **Growfinix Team.**

This internship report is submitted in partial fulfillment of the internship program requirements.

Date:

Growfinix Team

Student Signature

G. INDU

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to Growfinix for providing me the opportunity to undergo the internship on "C AND C++ DEVELOPMENT".

I am thankful to the Growfinix Team for their valuable guidance, support, and encouragement throughout the internship.

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G. INDU

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TASK 1 – EMPLOYEE RECORD MANAGEMENT SYSTEM USING C LANGUAGE

1.1 INTRODUCTION

Employee Record Management System is a console-based application developed using C programming language. The main purpose of this project is to maintain employee details efficiently in an organization.

This system helps in storing employee information such as Employee ID, Employee Name, Salary, Department, and Contact Details. The project provides functionalities to add, display, search, update, and delete employee records.

The project uses structures, functions, arrays, and file handling concepts of C language. It reduces manual work and helps in maintaining employee data in an organized manner.

The application is menu-driven and easy to use for beginners.

1.2 OBJECTIVE

The main objective of this project is to develop an Employee Record Management System using C language for storing and managing employee information efficiently.

The project aims to:
•Reduce manual work in managing employee records
•Store employee details systematically
•Provide easy access to employee information
•Implement CRUD operations (Create, Read, Update, Delete)
• Demonstrate file handling and structures in C language

1.3 SOFTWARE REQUIREMENTS

- Programming Language: C++
- IDE: Visual Studio Code
- Compiler: GCC/G++
- Operating System: Windows 10/11
- Storage: Text File/File Handling

1.4 SYSTEM DESIGN

The Employee Record Management System consists of the following modules:

1. Add Employee Record
2. View Employee Records
3. Search Employee Record
4. Update Employee Information
5. Delete Employee Record
6. Exit System

The system stores employee details using classes and file handling techniques. Each employee is assigned a unique Employee ID for identification.

1.5 FEATURES

- Add new employee records.
- Display all employee details.
- Search employees using Employee ID.
- Modify employee information.
- Delete unwanted records.
- Store records permanently using files.

1.6 ADVANTAGES

- Easy to use.
- Reduces paperwork.
- Improves data accuracy.
- Saves time and effort.
- Secure and organized record management

1.7 CONCLUSION

The Employee Record Management System successfully manages employee information using C++ programming concepts. The project demonstrates file handling, data storage, and object-oriented programming techniques. It provides an efficient solution for maintaining employee records and can be further enhanced with a graphical user interface and database integration.

TASK 2: BANK ATM SIMULATION

2.1 Introduction

The Bank ATM Simulation is a console-based application developed using C++. The project simulates the basic operations performed through an Automated Teller Machine (ATM). It enables users to perform banking transactions such as checking account balance, depositing money, withdrawing cash, and viewing account details.

The project demonstrates the practical implementation of Object-Oriented Programming concepts and provides a better understanding of banking transaction systems. It helps users learn how ATM systems operate in real-world banking environments.

2.2 Objectives

- To simulate the working of an ATM system.
- To perform banking transactions securely.
- To manage account balance efficiently.
- To implement menu-driven programming.
- To understand real-world banking operations.

2.3 Software Requirements

- Programming Language: C++
- IDE: Visual Studio Code
- Compiler: GCC/G++
- Operating System: Windows 10/11

2.4 System Design

The system consists of the following modules:

1. Check Balance
2. Deposit Amount
3. Withdraw Amount
4. Display Account Details
5. Exit

The user interacts with the system through a menu-driven interface and performs various banking operations.

2.5 Features

- Balance inquiry.
- Cash withdrawal.
- Cash deposit.
- Account information display.
- User-friendly interface.

2.6 Advantages

- Easy banking simulation.
- Fast transaction processing.
- Better understanding of ATM operations.
- Improves programming skills.

2.7 Conclusion

The Bank ATM Simulation project successfully demonstrates the implementation of banking operations using C++. It provides practical exposure to menu-driven applications and transaction management systems.

TASK 3: INVENTORY MANAGEMENT SYSTEM

3.1 Introduction

The Inventory Management System is a software application developed using C++ to manage and track product inventory efficiently. The system helps businesses maintain records of products, monitor stock levels, and update inventory information.

Inventory management plays an important role in ensuring smooth business operations. This project simplifies inventory control and reduces the chances of stock-related errors.

3.2 Objectives

- To maintain product inventory records.
- To track stock availability.
- To add, update, and delete product information.
- To reduce inventory management errors.
- To improve inventory organization.

3.3 Software Requirements

- Programming Language: C++
- IDE: Visual Studio Code
- Compiler: GCC/G++
- Operating System: Windows 10/11

3.4 System Design

The system contains the following modules:

1. Add Product
2. Display Products
3. Search Product
4. Update Product Details
5. Delete Product
6. Exit

Each product is assigned a unique Product ID for identification.

3.5 Features

- Product management.
- Stock monitoring.
- Product search facility.
- Inventory updating.
- Record deletion.

3.6 Advantages

- Accurate inventory tracking.
- Reduces manual effort.
- Easy product management.
- Improves business efficiency.

3.7 Conclusion

The Inventory Management System provides an efficient solution for maintaining inventory records. It demonstrates the application of C++ programming and file handling techniques in inventory control systems.

TASK 4: STUDENT GRADING SYSTEM

4.1 Introduction

The Student Grading System is a C++ application designed to calculate student grades based on academic performance. The system accepts marks obtained by students in various subjects, calculates percentages, and assigns grades according to predefined criteria.

The project helps educational institutions manage student performance effectively and ensures accurate grade calculation.

4.2 Objectives

- To calculate student percentages.
- To assign grades automatically.
- To maintain student academic records.
- To reduce errors in grading.
- To automate result processing.

4.3 Software Requirements

- Programming Language: C++
- IDE: Visual Studio Code
- Compiler: GCC/G++
- Operating System: Windows 10/11

4.4 System Design

The system performs the following operations:

1. Enter Student Details
2. Enter Subject Marks
3. Calculate Percentage
4. Generate Grade
5. Display Result

Grades are assigned based on the percentage obtained by students.

4.5 Features

- Student information management.
- Percentage calculation.
- Automatic grade generation.
- Result display.

- Easy-to-use interface.

4.6 Advantages

- Accurate grading.
- Saves evaluation time.
- Reduces manual errors.
- Efficient result processing.

4.7 Conclusion

The Student Grading System successfully automates the process of calculating student grades and percentages. It demonstrates the use of conditional statements, classes, and data management in C++.

TASK 4: TO-DO TASK MANAGER

5.1 Introduction

The To-Do Task Manager is a console-based application developed using C++ that helps users organize and manage daily tasks efficiently. The system allows users to add, view, update, and delete tasks. It improves productivity by helping users keep track of pending and completed activities.

5.2 Objectives

- To manage daily tasks effectively.
- To organize work schedules.
- To add and remove tasks easily.
- To track completed and pending tasks.
- To improve productivity and time management.

5.3 Software Requirements

- Programming Language: C++
- IDE: Visual Studio Code
- Compiler: GCC/G++
- Operating System: Windows 10/11

5.4 System Design

1. Add Task
2. View Tasks
3. Update Task
4. Delete Task
5. Mark Task as Completed
6. Exit

5.5 Features

- Task creation.
- Task updating.
- Task deletion.
- Task completion tracking.
- User-friendly menu interface.

5.6 Advantages

- Better task organization.
- Improved productivity.

- Easy task tracking.
- Simple and efficient operation.

5.7 Conclusion

The To-Do Task Manager project provides a simple and effective solution for managing daily activities. It demonstrates the practical implementation of data management concepts and helps users improve task organization and productivity.

After these tasks, add a final **Conclusion** page summarizing all five projects and the skills learned (C++, OOP, file handling, menu-driven programming, and problem-solving). This will make your Growfinix internship document look complete and professional.

CONCLUSION

The Growfinix Internship provided an excellent opportunity to enhance my programming and problem-solving skills through practical project development.

During the internship, I successfully completed five projects: Employee Record Management System, Bank ATM Simulation, Inventory Management System, Student Grading System, and To-Do Task Manager.

These projects helped me gain hands-on experience in C++ programming, Object-Oriented Programming (OOP), file handling, data management, and menu-driven application development. I learned how to design, implement, test, and document software projects effectively.

Overall, this internship improved my technical knowledge, logical thinking, and project development skills, preparing me for future academic and professional challenges in software development.